Line-by-Line Explanation of the Maple Code for the Key Vanishing Lemma (Compact Case)

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Degree 17, $m = 5$, Twist $= -5$

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1 Introduction

This Maple code proves a special case of the Key Vanishing Lemma for smooth surfaces in $\mathbb{P}^3$ (compact case). It verifies that for a generic surface $X \subset \mathbb{P}^3$ of degree $d = 17$ there are no nonzero invariant 2-jet differentials of weighted order $m = 5$ twisted by $\mathcal{O}_X(-5)$. The parameters are

$$m = 5, \qquad \text{twist} = -5, \qquad d_{\text{aux}} = 16 \quad \text{(so the surface degree is } d_{\text{aux}} + 1 = 17\text{)}.$$

The algorithm enforces holomorphicity conditions that a global section must satisfy, first on the affine charts and then across the hyperplane at infinity. By solving large linear systems, all coefficients of a candidate differential are forced to zero. By semicontinuity, the vanishing extends to a generic surface of the same degree.

1.1 Flexibility of the parameters

The three essential numbers are set at the very top:

`m`, `twist`, `d`.

For a surface of degree D one takes `d` $= D - 1$. By adjusting m, twist and d, the same code verifies all required cases. The present run with $m = 5$, twist $= -5$, $d = 16$ is a concrete illustration.

1.2 Limitations

The same construction, with the appropriate choice of the auxiliary monomial in the surface equation, should in principle prove the key vanishing lemma for other degrees as well. However, the resulting linear systems become too large for Maple to handle in a reasonable time for certain parameter regimes. It is very likely that a translation of the algorithm into C++ combined with parallel computation would overcome this obstacle, but this is beyond the scope of the present Maple implementation.

Now we proceed line by line through the Maple code.

2 Initialization

```
restart: m := 5; twist := -5; d := 16; degree := d + 1;
```

`restart:` clears all previous definitions.
`m := 5:` weighted order of the invariant 2-jet differential.
`twist := -5:` the bundle is tensored with $\mathcal{O}_X(-5)$.
`d := 16:` auxiliary integer; the true surface degree is `degree` $= d + 1 = 17$.

3 The surface equation

```
HUU := X^(d+1) + T^(d+1) + X^(floor((d+1)/2)) * T^(d+1-floor((d+1)/2)):
UU := subs({T=1, X=x}, HUU);
```

`HUU` defines the $X - T$ part of the homogeneous equation

$$\hat{R}(T, X, Y, Z) = X^{17} + T^{17} + X^8T^9 + Y^{17} + Z^{17}.$$

Here `floor((d+1)/2)` $= 8$, so the extra monomial is X^8T^9.

`UU` evaluates at $T = 1$, giving $x^{17} + 1 + x^8$. This will be part of the affine equation $R(x, y, z) = 0$.

4 Template of the invariant 2-jet differential

```
Jetxy := expand(
  (1/Rz^(m-0*2)) * add(AA||j * x1^(m-0*3-j) * (y1^j * Delta_yx^0), j=0..m-0*3) +
  (1/Rz^(m-1*2)) * add(BB||j * x1^(m-1*3-j) * (y1^j * Delta_yx^1), j=0..m-1*3) +
  (1/Rz^(m-2*2)) * add(CC||j * x1^(m-2*3-j) * (y1^j * Delta_yx^2), j=0..m-2*3) +
  (1/Rz^(m-3*2)) * add(DD||j * x1^(m-3*3-j) * (y1^j * Delta_yx^3), j=0..m-3*3)
):
```

This constructs the general local form of an invariant 2-jet differential on the chart $\{R_z \neq 0\}$ (Proposition 5.1):

$$J_{yx} = \sum_{k=0}^{\lfloor m/3 \rfloor} \frac{1}{R_z^{m-2k}} \sum_{j=0}^{m-3k} F_{j,k}(x, y, z)(y')^j (x')^{m-3k-j} (y'x'' - y''x')^k.$$

Here `x1`, `x2` denote x', x''; `y1`, `y2` denote y', y''; `Delta_yx` is the Wronskian $y'x'' - y''x'$; and `AA||j`, `BB||j`, `CC||j`, `DD||j` are undetermined polynomials.

5 Degree bounds for the coefficient polynomials

```
dA := (m-0*2)*d - m + twist:
dB := (m-1*2)*d - m + twist:
dC := (m-2*2)*d - m + twist:
dD := (m-3*2)*d - m + twist:
```

Each coefficient $F_{j,k}$ (labelled $\texttt{AA}_j$, $\texttt{BB}_j$, $\texttt{CC}_j$, $\texttt{DD}_j$) will be a polynomial with total degree in x, y, z bounded by the values `dA`, `dB`, `dC`, `dD`. These bounds come from the homogeneous degree constraints described in the paper.

6 Wronskian and change of chart formulas

```
Delta_yx := y1*x2 - y2*x1:
Sol1 := - x1*Rx/Ry - z1*Rz/Ry:
Sol2 := - x1^2*Rxx/Ry + 2*x1^2*Rxy*Rxy*Rz/Ry^2 - 2*x1*z1*Rxx/Ry
        - Ryy*x1^2*Rxx/2/Ry^3 - 2*Ryy*x1*Rxx*z1*Rz/Ry^3
        - Ryy*z1^2*Rz^2/Ry^3 + 2*z1*Rxy*x1*Rxy*2 + 2*z1*Ryz*Rz/Ry^2
        - z1^2*Rzz/Ry - x2*Rx/Ry - z2*Rz/Ry:
```

`Delta_yx` is the Wronskian $y'x'' - y''x'$. `Sol1` gives y' from the relation $x'R_x + y'R_y + z'R_z = 0$. `Sol2` gives y'' after substituting y' into the second derivative relation.

7 Transition to the other affine chart $(R_y \neq 0)$

```
Jetzx := sort(mtaylor(
  expand(subs(x2 = (Deltazz + x1*z2)/z1,
         expand(subs({y1 = Sol1, y2 = Sol2}, Jetxy)))),
  [x1, z1, Deltazz], 100), [x1, z1, Deltazz]):
```

We substitute `y1`, `y2` and replace `x2` using the relation

$$x_2 = (\Delta_{zx} + x_1 z_2)/z_1$$

obtained from the Wronskian $\Delta_{zx} = z'x'' - z''x'$. The result is expanded as a Taylor series in x', z', Δ_{zx} up to order 100. This yields the expression of the jet differential on the chart $\{R_y \neq 0\}$.

8 Extract coefficients after clearing denominators

```
printflevl := 2:
for k from 0 to m/3 do
  for j from 0 to m-3*k do
    EEEq[m-j-3*k,j,k] := factor(
      Rz^(m-j-3*k) * Ry^(m-2*k) *
      coeftayl(Jetzx, [x1,z1,Deltazz] = [0,0,0], [m-j-3*k,j,k])
    ):
  od:
od:
```

For each triple (i, j, k) with $i+j+3k = m$, we extract the coefficient of $(x')^i(z')^j(\Delta_{zx})^k$ and multiply by $R_z^i R_y^{m-2k}$ to clear allowed denominators. The result `EEEq[i,j,k]` is a polynomial that must satisfy divisibility conditions by R_z in the quotient ring $\mathfrak{R} = \mathbb{C}[x, y, z]/(R)$.

9 Define the explicit surface and its partial derivatives

```
RR := z^(d+1) + y^(d+1) + UU:
```

```
Rx := diff(RR,x): Ry := diff(RR,y): Rz := diff(RR,z):
Rxx := diff(RR,x,x): Rxy := diff(RR,x,y): Rxz := diff(RR,x,z):
Ryy := diff(RR,y,y): Ryz := diff(RR,y,z): Rzz := diff(RR,z,z):
```

The affine equation is

$$R(x, y, z) = z^{17} + y^{17} + x^{17} + x^8 + 1.$$

All first and second order partial derivatives are computed symbolically.

10 Truncation in z and preparation for y-reduction

```
printflevl := 2:
for k from 0 to m/3 do
  for j from 0 to m-3*k do
    EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], z, (m-j-3*k)*d):
  od:
od:
```

Because the divisibility condition requires that each `EEEq` be divisible by $z^{(d-1)i} = z^{15i}$ (since $R_z = 17z^{16}$), we truncate the Taylor expansion in z at order $i \cdot d$ (where $i = m - j - 3k$). This truncation will later be used to collect vanishing coefficients.

11 Reduction of high y-powers using the surface equation

```
for j1 from 0 to 1+m*d+d do
  ny[j1] := (-z^(d+1) - UU)^(floor(j1/(d+1))) *
            y^(j1 - (d+1)*floor(j1/(d+1))):
od:
```

In the quotient ring $\mathfrak{R}$, the relation

$$y^{d+1} = -z^{d+1} - x^{d+1} - x^8 - 1$$

allows us to replace any monomial y^{j_1} (with $j_1 \geq d + 1$) by a polynomial of y-degree $< d + 1$. The array `ny[j1]` gives the reduced form of y^{j_1}: for $j_1 < d + 1$ it is simply y^{j_1}; otherwise we factor out $(y^{d+1})^q$ and substitute the right-hand side. This implements Lemma 5.2 of the paper.

12 Expand the unknown polynomials as functions of y only (first stage)

```
for l from 0 to m-0*3 do
  AA||l := add(A||l[j]*y^j, j=0..min(d,dA)):
od:
for l from 0 to m-1*3 do
```

```
  BB||l := add(B||l[j]*y^j, j=0..min(d,dB)):
od:
for l from 0 to m-2*3 do
  CC||l := add(C||l[j]*y^j, j=0..min(d,dC)):
od:
for l from 0 to m-3*3 do
  DD||l := add(D||l[j]*y^j, j=0..min(d,dD)):
od:
```

At this stage the coefficients are written as polynomials in y alone (their coefficients are still undetermined functions of x and z). The degree in y is bounded by $\min(d, \text{total degree bound})$.

13 Reduce the equations in y-degree and extract coefficients

```
printflevl := 2:
for k from 0 to m/3 do
  for j from 0 to m-3*k do
    degree(EEEq[m-j-3*k,j,k], y):
    EEEq[m-j-3*k,j,k] := mtaylor(EEEq[m-j-3*k,j,k], y, 1+m*d+d):
    dy[m-j-3*k,j,k] := degree(EEEq[m-j-3*k,j,k], y):
  od:
od:
```

We expand each `EEEq` in y up to a sufficiently high order and record the resulting y-degree. Then for each power of y we extract its coefficient and substitute the reduction `ny[j1]`.

14 Further expansion in x and z

```
printflevl := 3:
for l from 0 to m-0*3 do
  for j from 0 to min(d,dA) do
    for k from 0 to dA-j do
      A||l[j,k] := add(A||l[i,j,k]*x^i, i=0..dA-j-k):
    od:
  od:
od:
```

Similar blocks exist for `BB`, `CC`, and `DD`. At this stage, the unknown coefficients are expanded as full polynomials in x, y, and z with undetermined constant coefficients.

15 Define the final truncated equations Eq

```
printflevl := 2:
for k from 0 to m/3 do
```

```
  for j from 0 to m-3*k do
    Eq[m-j-3*k,j,k] := mtaylor(Eq[m-j-3*k,j,k], z, (m-j-3*k)*d):
    degree(Eq[m-j-3*k,j,k], z):
  od:
od:
```

`Eq[i,j,k]` is the polynomial that must be divisible by $z^{(d-1)i}$; all its coefficients of $z^0, \ldots, z^{i(d-1)-1}$ must vanish.

16 Iterative solving of divisibility by $z^{\ell d}$

The core computation is an induction on $\ell = 1, 2, \ldots, 11$. At step ℓ, we require that all $\mathtt{Eq}[i, j, k]$ with $i = \ell$ be divisible by $z^{\ell d}$. We collect the coefficients of monomials $x^p y^q z^r$ with $r < \ell d$ and set them to zero. The resulting linear system is solved and the solutions are assigned globally.

16.1 First iteration ($\ell = 1$): divisibility by $z^{1 \cdot d}$

```
Systemez1d := {}:
for k from 0 to (m-1)/3 do
  Coeffs_z1d[1,m-1-3*k,k] := {coeffs(expand(mtaylor(Eq[1,m-1-3*k,k], z, 1*d)), [x,y,z
  Systemez1d := Systemez1d union Coeffs_z1d[1,m-1-3*k,k]:
od:
Variables_z1d := indets(Systemez1d):
Solutions_z1d := solve(Systemez1d, Variables_z1d):
assign(Solutions_z1d):
```

After solving, the number of terms in each coefficient is printed. For $m = 5$, the output shows:

```
AA0: 34476, AA1: 34476, AA2: 34476, AA3: 34476, AA4: 150700, AA5: 269140
```

and similarly for `BB`, `CC`, `DD`. The same pattern is repeated for $\ell = 2, 3, \ldots, 10, 11$. At each step the systems get larger and the number of free parameters decreases. By the end of the 11th iteration, all coefficients have been forced to zero.

16.2 Subsequent iterations

Similar blocks exist for $\ell = 2, 3, \ldots, 11$. For example, for $\ell = 4$:

```
Systemez4d := {}:
for k from 0 to (m-4)/3 do
  Coeffs_z4d[4,m-4-3*k,k] := {coeffs(expand(mtaylor(Eq[4,m-4-3*k,k], z, 4*d)), [x,y,z
  Systemez4d := Systemez4d union Coeffs_z4d[4,m-4-3*k,k]:
od:
Variables_z4d := indets(Systemez4d):
Solutions_z4d := solve(Systemez4d, Variables_z4d):
assign(Solutions_z4d):
```

The number of terms after each iteration is printed. By the end of the 11th iteration, all coefficients have been forced to zero, confirming the vanishing result.

17 Holomorphicity at infinity

After the affine divisibility conditions are satisfied, we must check that the differential extends holomorphically across the divisor at infinity $\{T = 0\}$.

17.1 Homogenisation of the coefficients

```
for l from 0 to m-0*3 do
  HA||l := add(add(add(A||l[i,j,k]*X^i*Z^k*Y^j*T^(dA-i-j-k),
               i=0..dA-j-k), k=0..dA-j), j=0..min(d,dA)):
od:
```

Each coefficient is homogenised to a homogeneous polynomial in T, X, Y, Z of degree `dA`, `dB`, etc. The results are stored in `HA||l`, `HB||l`, etc.

17.2 Form the combinations appearing in the transition formula

For the transition to the chart at infinity, we first extract coefficients from the homogenised forms:

```
for h from 0 to m-3*0 do
  for j1 from 0 to dYA[h] do
    CoeffA[h,j1] := coeff(HAEEq[h], Y, j1):
  od:
od:
```

Then we apply the reduction using `nY[j1]` and expand in T:

```
for h from 0 to m-3*0 do
  HAEq[h] := expand(mtaylor(add(CoeffA[h,j1]*nY[j1], j1=0..dYA[h]), T, m-3*0-h)):
  degree(HAEq[h],Y):
od:
```

The analogous constructions are performed for `HBEq[h]`, `HCEq[h]`, and `HDEq[h]`. This ensures that:

1. The coefficients are correctly extracted from the homogeneous polynomials `HAEEq[h]` as functions of Y;
2. The reduction `nY[j1]` (which encodes the relation $Y^{d+1} = -Z^{d+1} - X^{d+1} - X^8T^9 - T^{d+1}$) is applied correctly;
3. The Taylor expansion in T (the coordinate vanishing at infinity) is performed to the appropriate order $m - 3k - h$.

17.3 Verification of holomorphicity at infinity

```
Liste_x := {x = 1/x, y = y/x, z = z/x,
            x1 = -x1/x^2, x2 = 2*x1^2/x^3 - x2/x^2,
            y1 = y1/x - y*x1/x^2,
            y2 = y2/x - 2*y1*x1/x^2 + 2*y*x1^2/x^3 - y*x2/x^2}:
```

This transformation maps the chart at infinity to a neighbourhood of the origin. After applying this transformation, the differential must have no poles. The code verifies this by checking that all coefficients of negative powers vanish, confirming that the differential extends holomorphically across the divisor at infinity.

18 Conclusion

The Maple code successfully verifies that for a generic surface of degree 17, there are no nonzero invariant 2-jet differentials of weighted order $m = 5$ twisted by $\mathcal{O}_X(-5)$. By semicontinuity, the vanishing holds for a generic surface of the same degree.